

RICCARDO PUGLIESE

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Education

Sapienza University of Rome

MSc in Data Science

Rome, Italy

September 2025 - September 2027

- **Selected Coursework:** Fundamentals of Data Science; Algorithmic Methods of Data Mining and Laboratory; Fundamentals of Statistical Learning; Statistical Machine Learning; Data Management for Data Science; Natural Language Processing and Text Mining; Data Privacy and Security.

Università degli Studi Roma Tre

BSc in Economics and Big Data

Rome, Italy

October 2022 - March 2025

- GPA: 4.0, cum laude; Minor in Data Science
- **Coursework:** Machine Learning and Data Processing, Algorithms, Statistics for Big Data.

Université libre de Bruxelles

Erasmus+ Programme

Bruxelles, Belgium

August 2026 - January 2027

- **Coursework:** Cryptology, Database System Architecture, Data Mining.

Mykolas Romeris University

Erasmus+ Programme

Vilnius, Lithuania

August 2023 - January 2024

- **Coursework:** Databases, Computer Network Architectures, Business Mathematics.

Languages

Italian

Listening: Advanced Writing: Advanced Speaking: Advanced Reading: Advanced

Mother tongue

English

Listening: Advanced Writing: Advanced Speaking: Advanced Reading: Advanced

Advanced

French

Listening: Beginner Writing: Beginner Speaking: Beginner Reading: Beginner

Beginner

Research & Projects

Undergraduate Thesis: Strategic Innovative Investments in Business

- Reviewed literature on strategic and innovative investments to assess how innovative investments can be classified within strategic-investment categories, focusing on defining characteristics and competitive advantage.
- Proposed a **case-by-case, similarity-based approach** that assigns each innovative investment to the strategic category with the closest mechanism for creating competitive advantage.

DreamBank Enrichment and Dream Continuation Generation | Python, Qwen3, QLoRA

- Collaboratively enriched **21,000 DreamBank reports** with RoBERTa-based emotion labels, 25 BERTopic themes, and demographic metadata; in a four-model comparison, **QLoRA fine-tuning** lowered token-level NLL and perplexity, while metadata added little.

Bayesian Item Response Theory on INVALSI Data | R, JAGS

- Implemented Bayesian **Rasch/1PL and 2PL models** for 3,500 students across 49 Grade 10 INVALSI Italian items; 2PL reduced log loss (**0.518 vs. 0.527**), AIC, BIC, and DIC, while ability estimates closely matched official WLE scores (**r = 0.990**).

Pokémon Battle Outcome Prediction | Python, scikit-learn, XGBoost

- In a three-person team, engineered features from 10,000 JSONL battle logs and tuned logistic regression, AdaBoost, and XGBoost; a **soft-voting ensemble** achieved **0.846 public-leaderboard accuracy and second place**.

Vessel-Type Classification from AIS Data | Python, TensorFlow/Keras

- Collaboratively developed a residual feed-forward neural network to classify **17 vessel macro-types** from NOAA AIS-derived geometry and speed features; hybrid resampling and class-weighted loss achieved **0.8149 test accuracy and 0.8028 macro F1**.

Conformal Prediction Bands for Heart-Rate Curves | R, FPCA, GMM

- Transformed approximately 13 months of Samsung Health measurements into **148 daytime and 159 nighttime functional curves**, then combined FPCA, a two-component Gaussian Mixture Model, and projection-based conformal inference to construct **90% prediction bands** for future heart-rate curves.

Research & Projects - Continued

Home Credit Relational Database | PostgreSQL, SQL, Python

- Collaboratively designed a PostgreSQL schema for eight Home Credit source files plus a unified parent table; built a chunked PyArrow/SQLAlchemy loader, wrote **10 analytical queries**, and optimized joins using indexes and **EXPLAIN (ANALYZE, BUFFERS)**.

PROGRESA Eligibility and Migration | R, Difference-in-Differences

- In a five-person team, estimated the effect of a PROGRESA eligibility proxy on first-time U.S. migration using Mesoamerican Migration Project data, **Difference-in-Differences** and **an event study**; estimates were **near zero and statistically insignificant** across specifications.

Italian University Sustainability and Territorial Gaps | Ontology, Regression, Vega-Lite

- Integrated ranking and institutional data for Italian universities, extended a sustainability ontology, and analysed **58 universities** through regression and visual analytics; results indicated a **positive sustainability-reputation association** and a North-Central versus South territorial gap.

Unsupervised Book Clustering | Python, TF-IDF, NMF, KMeans

- Parsed **1,934 book HTML pages** into a Parquet dataset, generated 43,918 TF-IDF features, learned 20 NMF topics, and grouped the books into **15 KMeans clusters** for the final submission.

South African Marriage and Cohabitation Analysis | R, Statistical Modelling

- Analysed survey data covering **40,944 South African individuals**, cleaned and reconstructed marital-status information with logistic regression (**AUC = 0.833**), and fitted multiple linear models to examine factors associated with marriage and cohabitation duration.

IDENTITY: Blockchain-Based GDPR Compliance Proposal | Hedera, Smart Contracts, GDPR

- In a five-person team, designed a B2B proposal for encrypted on-chain personal-data storage and auditable access through Hedera smart contracts; assessed GDPR Article 17 feasibility, the business model, SWOT factors, and technical trade-offs.

Technical Skills

Programming: Python, R, SQL

Machine Learning & Statistical Modelling: scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, Hugging Face Transformers, PEFT/QLoRA, BERTopic, JAGS/rjags, FPCA, Gaussian Mixture Models, conformal prediction

Data Processing & Visualisation: pandas, NumPy, PyArrow, BeautifulSoup, Matplotlib, ggplot2, Vega-Lite

Databases: PostgreSQL, SQLAlchemy, pgAdmin

Tools: Git, Jupyter Notebook, LaTeX

All projects, along with my exam transcripts and complete study plans, are available on my portfolio and GitHub:

<https://1fiocco.github.io/> | <https://github.com/1fiocco>